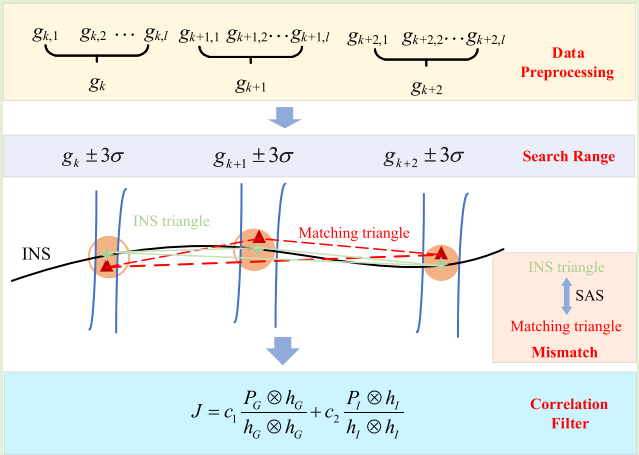
# Gravity Matching Algorithm Based on Correlation Filter

Shengwu Zhao, Xuan Xiao, Zhihong Deng, and Lei Shi

***Abstract*—The gravity-matching algorithm is one of the key technologies in the gravity-aided inertial navigation system (GAINS). The matching accuracy determines the correction accuracy of the inertial navigation system (INS). However, the initial position error of the INS and gravity measurement error lead to a decrease in matching accuracy. In this article, the characteristics of gravity measurement error and inertial navigation information are made full use of to reduce the impact on matching. A gravity-matching algorithm based on a correlation filter (CF) is proposed, which includes preprocessing, CF, and mismatch detection. Meanwhile, the shape of the INS trajectory is used as a constraint to reduce the mismatch caused by the measurement error to improve the matching accuracy. Moreover, two matching strategies are given, including normal matching and sliding window matching. Experimental results show that the proposed method can effectively improve the matching accuracy under initial position error and gravity measurement error.**




***Index Terms*—Correlation filter (CF), gravity-aided navigation, inertial navigation system (INS) error, matching algorithm.**

## I. Introduction

With further exploration and development of the ocean, autonomous underwater vehicles (AUVs) as high-tech have become a hot spot in maritime technology and research. When AUVs work, navigation and positioning techniques are the keys to ensuring the completion of the task [1]. For long-term navigation, AUVs have the requirements of self-containment, a wide range of movement, and high concealment. Hence, an inertial navigation system (INS) has become a core navigation system. The INS can independently provide all-day, long-term navigation information without being interfered with by the external environment and has high concealment. However, the navigation information is obtained by integrating accelerometer and gyroscope data, which leads to the divergence of the position over time. Using geophysical parameters such as gravitational fields, geomagnetic fields, topography, and landforms can effectively improve the positioning accuracy of AUVs. The measurement of topography and geomorphology is limited by the range of sonar and optical cameras in the underwater situation. Geomagnetic measurement is easily affected by the environment. These factors restrict the use of these methods. The gravitational field generally does not change with time, has good temporal and spatial distribution characteristics, and the measurement technology is mature [2]. Therefore, gravity-aided INS (GAINS) has become a hot topic in underwater autonomous navigation.

In GAINS, the gravity-matching algorithm is one of the key techniques using the relationship between the measurement values of gravity anomaly and gravity anomaly map to obtain the position estimation information through certain criteria. Traditional gravity-matching methods mainly include the Sandia inertial terrain-aided navigation (SITAN) algorithm [3], iterated closest contour point algorithm (ICCP) [4], [5], and terrain contour matching (TERCOM) algorithm [6], [7]. SITAN is a single-point matching algorithm based on the extend Kalman filter (EKF) [8] with strong real-time performance. The observation model is fitted from the local gravity background map. Owing to the irregular distribution of gravitational fields and the resolution of the gravity anomaly map, it is difficult to obtain an accurate observation model, and the observation model needs to be linearized in the calculation process, which seriously affects the estimation accuracy [9], [10]. The particle filter (PF) [11], [12] is also

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The authors are with the School of Automation and the National Key Laboratory of Intelligent Control and Decision of Complex Systems, Beijing Institute of Technology, Beijing 100081, China (e-mail: zhaoshengwu@bit.edu.cn; bitxxuan@bit.edu.cn; dzh_deng@bit.edu.cn; bitshilei@gmail.com).

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used as a single-point matching algorithm which has higher accuracy than the EKF. Based on PF, a vector-matching algorithm [11] is proposed, overcoming the defect of the traditional single-point matching algorithm that diverges in some matching regions where gravity changes are not obvious. It still needs an accurate model. The ICCP algorithm is a sequence iterative algorithm. It does not need a model, and it has high accuracy but poor real-time performance. However, the ICCP algorithm is based on the assumption that the gravity anomaly measurement value is accurate and the initial position error is small. If these two conditions are not met, the ICCP algorithm has a poor matching result [13], [14]. Aiming at the matching problem under the condition of large initial position error, the artificial bee colony algorithm [14] and triangle matching algorithm [15], [16] based on affine transformation are proposed. The TERCOM algorithm is a typical sequence-matching algorithm. The algorithm achieves matching through point correlation analysis and is not affected by the initial position error. Nevertheless, the algorithm requires high sensor accuracy and its computational complexity is high, which leads to long calculation time and poor real-time performance. On the basis of the TERCOM algorithm, the shortest path algorithm is applied to increase the calculation frequency and calculation speed and make it a better real-time performance [13]. In addition, a matching algorithm based on the point mass filter (PMF) and the deterministic resampling method is proposed [17], which is able to achieve the positioning accuracy of gravity anomaly map resolution with sufficiently accurate measurement results. A matching algorithm that combines ICCP and PMF is proposed to achieve better results compared with the traditional ICCP algorithm [18].

Through the above analysis, the existing gravity-matching algorithms are either affected by the initial position error of INS, sensitive to the gravity measurement error, or both. When underwater vehicles enter the adaptation area, the initial position error of INS is usually unknown. Some gravity-matching algorithms may have low accuracy or even fail under large initial position errors. Sources of gravity measurement errors mainly include errors related to the measurement process of the gravimeter, errors caused by inaccurate correction of Utterfurth and inaccurate positioning. Combined with the influence of various errors in gravity measurement, which can be achieved the offshore is $\pm 1.0$–$\pm 2.0$ mGal. The accuracy is $\pm 1.5$–$\pm 3.0$ mGal in the ocean [19]. In addition, real-time performance is an important evaluation index of underwater navigation algorithms. Most sequence-matching algorithms (ICCP, TERCOM, etc.) do not have real-time performance, while single-point matching algorithms (SITAN, PF, PMF, etc.) do.

Hence, a gravity-matching algorithm based on the correlation filter (CF) [20] is proposed in this article aiming at the above problems. The proposed algorithm is less affected by the INS initial position error and gravity measurement error. In the proposed algorithm, data preprocessing is carried out to reduce the effect of the gravity measurement error and avoid the outlier. CF is improved under the background of GAINS. The shape of the INS trajectory is used as a constraint to reduce the mismatch caused by the measurement error to improve the matching accuracy. In addition, two matching strategies are given, including normal matching and sliding window matching. To be specific, the main contributions of this work can be concluded as follows.

1) CF is applied and improved in GAINS. A trajectory shape template and a gravity-matching template are proposed, which consider both the shape of INS and gravity measurement.
2) A novel mismatch algorithm based on triangle similarity is proposed, which uses the shape of the INS trajectory to correct the wrong matching points.
3) Strategies of normal matching and sliding window matching are given. Normal matching has a small amount of computation while sliding window matching not only has real-time performance but makes the matching result continuous.

The rest of the article is arranged as follows. In Section II, the basic principles of the algorithm are introduced. In Section III, a matching algorithm based on CF is proposed. Experimental results are shown in Section IV. Section V is the discussion. Section VI is the conclusion.

## II. Basic Principles of the Algorithm

Fig. 1 shows the block diagram of the matching algorithm using CF. First, doing data preprocessing to get rid of the noise from the measured gravity anomaly. Then determining the search range according to the characteristics of gravity measurement error and inertial navigation error to ensure that the real position is within this range. Designing the CF template is done according to the processed gravity anomaly data and inertial navigation information. Using CF for the template and the candidate matching points in the search range and determining the final position by correlation value. Finally, a triangle-based mismatch detection is performed to prevent a mismatch.

### A. Gravity Anomaly Data Preprocessing

*1) Content of the Method:* According to the introduction of gravity measurement in [19], we assume that the gravity measurement error is Gaussian white noise with the standard deviation $\sigma$. The gravity measurement data is $\alpha_1, \alpha_2, \ldots, \alpha_n$, where $n$ is the number of gravity samples at time $t_n$. When the sampling frequency of the gravimeter is 2 Hz, the distance traveled by the AUV in a short period of time is much less than the resolution of the gravity anomaly map. Then we assume that several continuous measurements are all the measurements of a fixed-point gravity anomaly. To reduce the influence of outliers on matching, a nonlinear filter method similar to the median filter is used as the preprocessing method to reduce the impact of impulse noise. First, take $m$ gravity measurement values around the time $t_n$, where $m$ can be obtained from the speed of the UAV and the resolution of the gravity anomaly map. Sort in ascending order $\alpha'_1, \alpha'_2, \ldots, \alpha'_m$. Then select $l$ gravity measurements in the middle as sample gravity measurement and the samples will be applied in template choosing. The process can be shown in Fig. 2.

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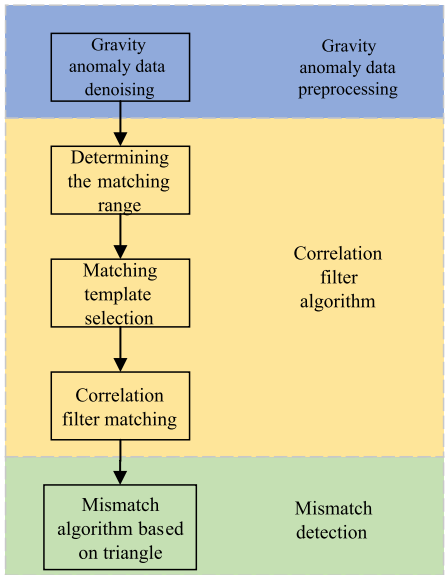


Fig. 1. Block diagram of the matching algorithm using CF.

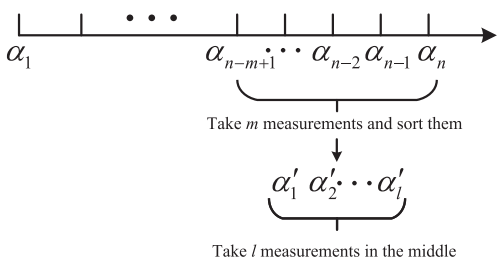


Fig. 2. Preprocessing process of gravity measurements.

Since this article assumes that the gravity measurement value within a certain time is unchanged, the setting of parameter $m$ should meet the following conditions:

$$m v_e \Delta g_e < 1852 \eta r_g \cos(L) f_g \sigma \tag{1}$$

$$m v_n \Delta g_n < 1852 \eta r_g f_g \sigma \tag{2}$$

where $v_e$ and $v_n$ are east and north velocities, respectively, $\Delta g_e$ and $\Delta g_n$ are the largest difference between the gravity anomaly values at two adjacent points in the gravity anomaly map in east and north direction, respectively, $f_g$ is the sampling frequency of gravity measurement, $r_g$ is the solution of gravity anomaly map, $\sigma$ is the standard deviation of measurements, $L$ is the latitude, and $\eta$ is a parameter that is less than 1. The meaning of the above inequalities is to guarantee that $m$ sampled gravity measurements can reflect the information about the true position at $\alpha_n$. That is, for the actual gravity anomaly values corresponding to $m$ positions, the difference between the maximum and the minimum is smaller than the standard deviation of measurements, and the distance traveled by underwater vehicles in $m$ time periods is less than the resolution of the map. This article sets $\eta = (1/3)$. When $m$ is determined, $l \approx (1/2)m$. Next, the rationality of the method will be explained.

*Remark 1:* In practice, the parameters in (1) and (2) cannot be accurately estimated. It is worth mentioning that (1) and (2) are only the conditions satisfied by $m$, which are equivalent to the upper limit of $m$. Obviously, due to the approximate estimation of various parameters, the value of $m$ should be less than the upper limit.

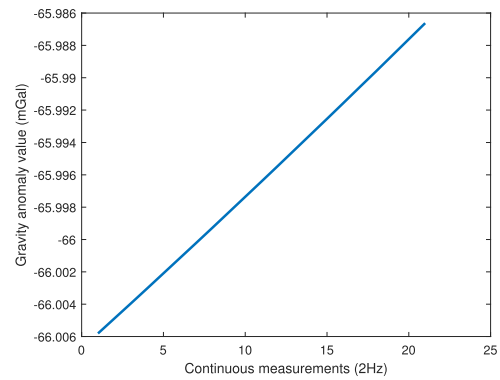


Fig. 3. Continuous gravity measurement values (In the adaptation area, the velocity is approximately equal to 3 m/s. The horizontal axis represents 21 consecutive samples under 2-Hz sampling frequency, and the vertical axis represents the corresponding gravity anomaly values).

*2) Rationality of the Method:* We assume that several continuous measurements are all the measurements of a fixed-point gravity anomaly. Strictly speaking, the true gravity anomaly values of these $l$ measurements are various. However, the difference in these measurements is much smaller than the error caused by measuring. A real measurement figure is shown in Fig. 3. The difference between the maximum and minimum gravity measurement values is 0.0192 mGal which can be ignored.

In addition, the article assumes that the measurement error is white noise with the standard deviation $\sigma$ and the errors at different times are independent of each other. By the above operation, it can be obtained that the gravity measurement error after data preprocessing still follows the Gaussian distribution with 0 mean and $(\sigma/\sqrt{m})$ standard deviation (central limit theorem). Hence, it can be seen that under Gaussian distribution, the accuracy can be improved. This is the principle of *taking the average of multiple measurements*. Although the actual measurement error may not satisfy the Gaussian distribution, the accuracy can be improved to some extent because the positive and negative errors cancel each other out. Meanwhile, the impulse noise or outlier can be dropped due to sorting. In fact, the proposed gravity matching algorithm based on CF does not have high requirements on the distribution characteristics of measurement errors, unless there are outliers in the measurement values. Therefore, the data preprocessing is significant which can ensure that the algorithm does not fail.

## B. Search Range Determination Method

In order to reduce the impact of measurement error further, the search range is expanded from the contour line to the contour belt, as shown in Fig. 4. The yellow line represents the contour line of the true gravity measurement, and the two black dashed lines represent the boundary of the contour belt.

In the contour belt shown in Fig. 4, when the width of the contour belt is $2\sigma$ in which $\sigma$ is the standard deviation

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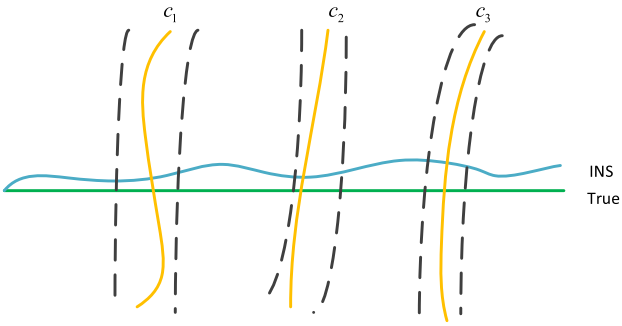


Fig. 4. Contour belt.

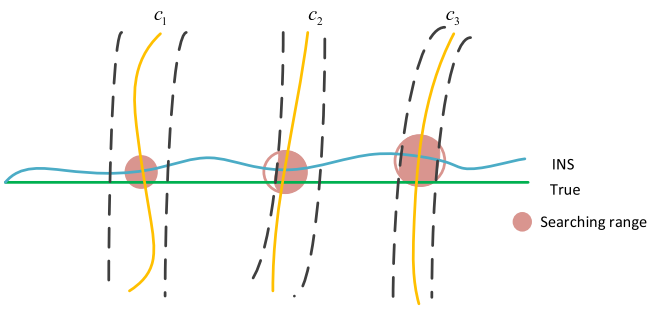


Fig. 5. Search range.

of gravity measurement error, the probability that the true position is in the contour belt is 0.9544. When the width of the contour belt is $3\sigma$, the probability is 0.9974. The calculation amount is larger than searching directly on the contour belt. Hence, the final range is determined by the inertial navigation information. The INS errors can be measured by the circular error probable (CEP). Yang et al. [15] and Park et al. [21] presented the relationship between the resolution, time, and scope of the gravity map. But it is only applicable to its own method and not universal. In this article, the final search range is the intersection of the contour belt and CEP, as shown in Fig. 5.

It is worth mentioning that due to the errors in the gravity anomaly map, INS, and so on, the obtained gravity measurement errors may not be Gaussian. Even with a Gaussian distribution, the variance may be unknown. At this time, the width of the contour belt can be set according to the actual situation and it is necessary to ensure that the real position has a high probability within the contour belt. For uncertain measurement error characteristics, the width can be set to a large value.

*Remark 2:* In practice, it might happen with a small chance that there is no intersection of the contour belt and CEP or the gravity anomaly map does not contain the search range. If so, the CEP range will be the final search range and match points will be corrected by the mismatch algorithm.

### C. CF Matching Algorithm

A CF matching algorithm is proposed in this section.

*1) CF Introduction:* CF was first applied to signal processing to describe the correlation between two signals. For two random signals $f$ and $h$, the correlation functions in continuous and discrete forms are defined as

$$\left(f \bigotimes h\right)(\tau) = \int_{-\infty}^{\infty} f^*(t)h(t+\tau)\, dt \tag{3}$$

$$\left(f \bigotimes h\right)(n) = \sum_{-\infty}^{\infty} f^*(m)h(m+n) \tag{4}$$

where $f^*$ represents the complex conjugate of $f$.

From the definition, the correlation function is similar to the convolution operation. The difference is that the two sequences of the correlation function are not flipped, and the sum is calculated by sliding and multiplying. However, a sequence in the convolution needs to be flipped first and done the sum. Therefore, the correlation function is equal to $f^*(-t)$ and $h(t)$ to do convolution. The intuitive interpretation of correlation is to measure the similarity of two signals at a certain moment. The more similar the two signals, the higher the correlation. From the perspective of linear space, correlation operations are actually inner product operations, and the inner product of two vectors in linear space represents the projection of one vector to another vector, indicating the similarity of the two vectors. Hence, correlation operations reflect this kind of similarity.

In gravity matching, we need to find the most similar trajectory in the search range as the matching result. According to this idea, we need to find a filter to maximize the response on the target, that is,

$$g = f \bigotimes h \tag{5}$$

where $g$ represents the response output (related output), $f$ represents input, and $h$ represents the filter template.

When we determine the filter template $h$, the response output is obtained. The calculation of (5) is to perform convolution calculations. This calculation on the computer takes a lot of time. When a fast Fourier transform (FFT) is performed on the above equation, it becomes an elementwise multiplication in the Fourier domain, which greatly reduces the amount of calculation. Equation (5) becomes the following form:

$$F(g) = F\left(f \bigotimes h\right) = F(f) \cdot F(h)^*. \tag{6}$$

For the convenience of description, write (6) as

$$G = F \cdot H^*. \tag{7}$$

Then the next step is to find $H^*$

$$H^* = \frac{G}{F}. \tag{8}$$

However, in the actual matching process, we must consider the influence of measurement errors and other factors and consider the target's trajectory as a reference to improve the robustness of the filter template. The minimum output sum of the squared error (MOSSE) model is

$$\min_{H^*} \sum_{i=1}^{m} \left|H^* F_i - G_i\right|^2. \tag{9}$$

The operations of (9) are all element-level, so as long as the MOSSE of each element ($w$ and $v$ are the indexes of each

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element in $H$) is the minimum. Therefore, (9) can be converted to

$$\min_{H^*_{wv}} \sum_{i=1}^{m} \left| H^*_{wv} F_{wvi} - G_{wvi} \right|^2. \tag{10}$$

To get the minimum $H^*_{wv}$, we only need to find the partial derivative of (10) and make the partial derivative 0

$$0 = \frac{\partial}{\partial H^*_{wv}} \sum_{i=1} \left| H^*_{wv} F_{wvi} - G_{wvi} \right|^2 \tag{11}$$

we have

$$H_{wv} = \frac{\sum_i F_{wvi} G^*_{wvi}}{\sum_i F_{wvi} F^*_{wvi}}. \tag{12}$$

The $H$ obtained by (12) is the value of each element, and finally $H$ is

$$H = \frac{\sum_i F_i G^*_i}{\sum_i F_i F^*_i} \tag{13}$$

or

$$H^* = \frac{\sum_i G_i F^*_i}{\sum_i F_i F^*_i}. \tag{14}$$

Then (13) or (14) is the model of the filter.

*2) Application in Gravity Matching:* CF algorithm carries out a correlation analysis between the template with real information and the input to be matched. In the gravity-matching process, for $N$ consecutive gravity matching points, assume that $p_k (k = 1, 2, \ldots, N)$ is a candidate position point at the $k$th gravity matching in the search range which is obtained in Section II-B. The candidate position point can be selected from the grid points within the search range in combination with the gravity anomaly map. The choice of the grid point lies in the fact that the gravity anomaly value can be read directly from the gravity anomaly map. Nongrid points are also needed to prevent restrictions on the direction of matching results.

Then the response can be calculated by

$$J = c_1 \left( P_G \bigotimes h_G \right) + c_2 \left( P_I \bigotimes h_I \right) \tag{15}$$

where $h_G$ and $h_I$ are the templates for gravity measurement and INS trajectory, respectively, $P_G$ and $P_I$ are the gravity anomaly sequence and candidate position point sequence to be matched, respectively, and $c_1, c_2$ are the weighting coefficients. The specific form of $P_I$ and $P_G$ is shown as

$$P_I = \begin{bmatrix} \mathrm{slo}(p_1, p_2) & \mathrm{slo}(p_2, p_3) & \cdots & \mathrm{slo}(p_{N-1}, p_N) \end{bmatrix} \tag{16}$$

$$P_G = \begin{bmatrix} h_m(p_1) & h_m(p_2) & \cdots & h_m(p_N) \end{bmatrix} \tag{17}$$

where $\mathrm{slo}(p_1, p_2)$ represents the slope of $p_1$ and $p_2$, and $h_m(p_1)$ represents the gravity anomaly value at position $p_1$ reading from gravity anomaly map.

The first half on the right considers the gravity measurement and finds the best candidate point sequence. The gravity anomaly values corresponding to points in this sequence are closest to the measurements. However, gravity information is not enough for precious position estimation because it does not include any position information. Therefore, the correlation of the inertial navigation trajectory and the trajectory composed of candidate positions is also used as one of the criteria for matching which is the last half on the right. When considering the trajectory of the inertial navigation, we only consider whether the shape is similar here. Calculate the slope of the inertial navigation trajectory points $(x_k, y_k)$ as a sequence and call it the trajectory shape template $h_I$. Next, we introduce how to obtain gravity template $h_G$.

For gravity matching point $p_k$, let $G_{ik}$ $(i = 1, 2, \ldots, l)$ represent the $i$th gravity anomaly value during the preprocessing of gravity data. Then, $g_i$ can be generated by

$$g_i = [K(1), K(2), \ldots, K(N)] \tag{18}$$

where

$$K(x) = \exp\left( -\frac{(x - x_c)^2}{2\sigma^2} \right) \tag{19}$$

and $x_c$ is the middle position of sequence $1{:}N$, that is, $x_c = (1+N)/2$. $\sigma$ is the stand deviation. In general, $\sigma$ is large since $N$ matching points have the same importance. $f_i$ is expressed by

$$f_i = [G_{i1}, G_{i2}, \ldots, G_{iN}]. \tag{20}$$

Therefore, $h_G$ can be obtained according to (13)

$$h_G = iF(H) = \frac{\Sigma_i F(f_i) F(g_i)^*}{\Sigma_i F(f_i) F(f_i)^*} \tag{21}$$

where $iF(H)$ represents the inverse FFT of $H$, or the iterative approach in [20].

*Remark 3:* The sizes of $f$, $g$, and $h$ are all the same here. From the measurement result, we find the response output $g_i$ has the characteristics of a large value in the middle and a small value around. We just need to find a function with the same characteristics and then we can get $g_i$ easily. The Gaussian function satisfies the characteristics. Hence, the Gaussian function is chosen to generate the response output.

Finally normalizing each response, we have

$$J = c_1 \frac{P_G \bigotimes h_G}{h_G \bigotimes h_G} + c_2 \frac{P_I \bigotimes h_I}{h_I \bigotimes h_I}. \tag{22}$$

When (22) is maximum, $P_I$ is the CF matching result.

*Remark 4:* After the template is determined, it is easy to get the maximum value. Since the number of candidate matching points is limited, the results can be obtained by matching the trajectories composed of all candidate points by an exhaustive method. The relationship of the position of the candidate points can be used as one of the constraints to optimize this process. $c_1, c_2$ can be determined by the accuracy of inertial and gravity sensors. Generally, $c_1 = c_2 = 0.5$.

## D. Mismatch Algorithm

Irregularities and uncertainties in the distribution of gravity anomaly lead to large differences in matching results. Sometimes, the errors of the matching points are larger than the inertial navigation errors. In order to solve this problem, a triangle method is proposed to determine the mismatch points.

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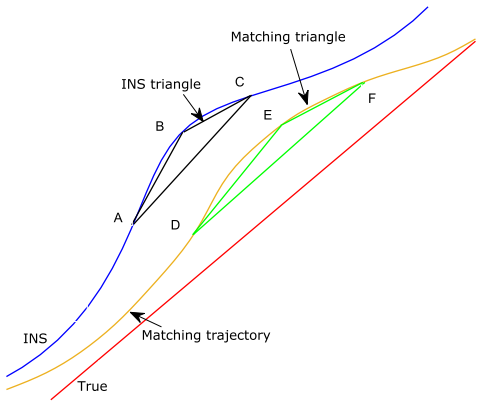


Fig. 6. INS triangle and matching triangle.

The triangle method uses the principle of triangle similarity criterion to detect mismatch points. It requires three consecutive INS trajectory points and three matching trajectory points to form an INS triangle and a matching triangle.

The judgment is based on the side angle side (SAS) theorem to determine how similar the two triangles are. The basic idea is shown in Fig. 6.

The calculation index can be expressed by the following equations:

$$\Delta\text{Ang} = |\angle B - \angle E| \tag{23}$$

$$\Delta\text{Ra} = \left|\frac{|AB|}{|BC|} - \frac{|DE|}{|EF|}\right| \tag{24}$$

where $\Delta$Ang is the angle index and $\Delta$Ra is the side length index.

Due to different errors, even if the three matching points of the triangle are completely correct, the INS triangle and the matching triangle cannot be completely similar. The purpose of this algorithm is to remove the points with larger errors, hence two triangles can be appropriately reduced the similarity

$$\Delta\text{Ang} < T_a \tag{25}$$

$$\Delta\text{Ra} < T_r \tag{26}$$

where $T_a$ represents the threshold of the angle difference, and $T_r$ is the side length difference. The point that does not satisfy (25) and (26) at the same time is a mismatch point.

Since the triangle method uses three consecutive INS trajectory points and three matching trajectory points, when a mismatch point occurs in the previous mismatch detection, the mismatch point cannot be used as the triangle in the next mismatch detection. Therefore, when a mismatch point appears, it is not only necessary to delete the mismatch point from the matching sequence, but also to construct a virtual matching point based on the characteristics of the matching triangle and the INS triangle. So the virtual matching point can be used in the next mismatch detection. If the point $F$ in Fig. 7 is a mismatch point, the virtual matching point $F_v$ is generated by

$$r = \frac{|AB|}{|DE|} \tag{27}$$

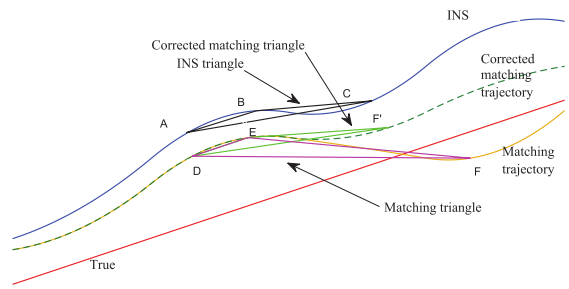


Fig. 7. Generation of mismatch points and virtual matching points.

$$x_{f1} = \frac{x_c - x_b}{r} + x_e \tag{28}$$

$$y_{f1} = \frac{y_c - y_b}{r} + y_e \tag{29}$$

$$x_{f2} = \frac{x_c - x_a}{r} + x_d \tag{30}$$

$$y_{f2} = \frac{y_c - y_a}{r} + y_d \tag{31}$$

$$x_{f_v} = \frac{x_{f1} + x_{f2}}{2} \tag{32}$$

$$y_{f_v} = \frac{y_{f1} + y_{f2}}{2} \tag{33}$$

where the coordinates of point $A$ are $(x_a, y_a)$, and the same for points $B$, $C$, $D$, $E$, $F$, $F_v$, $r$ is the similarity ratio of $|AB|$ and $|DE|$. $A$ and $B$ are the INS position points, $D$ and $E$ are the matching points. In order to generate the virtual point $F_v$, the similarity ratios of $|BC|$ and $|EF_v|$ and $|ac|$ and $|DF_v|$ should be the same. Then we can get two virtual points $F_1$ and $F_2$ by (28)–(31). The final virtual point $F_v$ is the mean of $F_1$ and $F_2$ according to (32) and (33).

There are some issues that need to be explained about this algorithm. As mentioned above, the error of the matching result may be larger than the error of the INS, which is regarded as the mismatch point. In other words, compared with the first two matching points, when the error of the third point to be evaluated increases significantly, this point is regarded as a mismatch point, which is similar to the occurrence of outliers in the measurement. This algorithm is not an optimization algorithm that further improves the accuracy of matching results through triangular similarity relations. It just tells us if it is a mismatch or not. Hence, the values of $T_a$ and $T_r$ are generally set to larger values. For an effective gravity matching algorithm, a mismatch should be a small probability event. In most cases, it is not necessary to generate a virtual point but to trust the results of the CF matching algorithm through (25) and (26), unless the matching results are very wrong. Therefore, even if the INS has a certain error, the results of the CF algorithm and mismatch algorithm will not have a great impact on the correction.

The mismatch algorithm proposed above is suitable for evaluating the third matching point when there are already two matching points. For the verification of the second matching point, the distance and direction relationship between the second matching point and the first matching point are calculated and compared with the distance and direction of the corresponding two position points of INS. If the point is

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TABLE I
GRAVITY MATCHING ALGORITHM USING CF

| Matching algorithm based on correlation filter |
|---|
| **Step 1** Preprocess the measured gravity data at each point to determine whether it needs to be matched (see Part II-A). If yes, proceed to step 2, otherwise proceed to step 7. |
| **Step 2** Determine the search range at each point (see Part II-B). |
| **Step 3** Calculate matching templates $h_G$ using (21) and $h_I$ using the slope of the INS points (see Part II-C.2). |
| **Step 4** For each candidate point from the search range, calculate the response $J$ using (22). |
| **Step 5** Output the maximum response sequence $P_I$ as the matching result. |
| **Step 6** Perform mismatch detection on the matching result using (25) and (26), and get the final matching result using (32) and (33) (see Part II-D). |
| **Step 7** Judge whether the voyage is over. If yes, it ends. Otherwise go to step 1. |

judged to be mismatched, the virtual point is generated by the shape of the INS trajectory.

For the verification of the first matching point, we assume that the first matching point is not a mismatch point and does not need correction in this article. The reason is that: according to the characteristic of the INS error, the search range proposed in Section II-B is not large compared with the latter matching points. The candidate matching points are limited in this search range, that is, most of the candidate matching points are not mismatch points. Hence, as long as the gravity measurement value does not have a large error, the error of the first matching point is not large, and it does not have a large impact on the detection of subsequent matching points. In fact, the first several matching points also have the characteristic.

*Remark 5:* According to the principle of the CF matching algorithm, the matching points are limited in the search range in Section II-B. In other words, when this range is not appropriate or very large, a mismatch may occur. In general, when the gravity measurement error is large or the underwater vehicle is in the nonadaptive area, it is easy to mismatch.

## III. ALGORITHM SUMMARY

For $N$ consecutive points to be matched, the CF matching algorithm can be divided into the following steps, as shown in Table I.

Similar to the ICCP algorithm, two matching strategies are given in this article, as shown in Fig. 8. In the first sampling method, gravity matching is performed at every $N$ matching point, and there is no coincidence between two adjacent gravity matching processes, which is called normal CF (N-CF). The second method is the sliding window method, in which two adjacent gravity-matching processes slide only one point, which is called sliding window CF (SW-CF). The first method requires less computation but lacks real-time performance. The second method is real-time, and the results

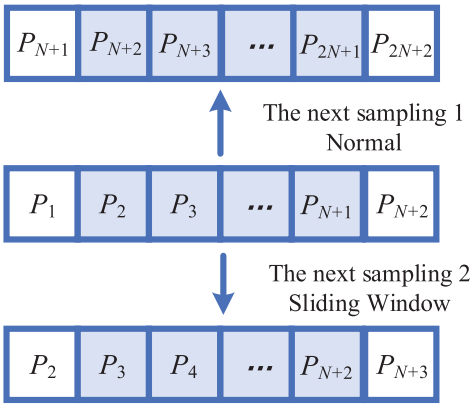


Fig. 8. Two matching strategies.

obtained are more continuous than the first method. Apart from the above, in SW-CF, a matching point can be obtained every time CF is processed, and this point can be combined with the CF processing of the last two times through the mismatch algorithm. So, continuity will improve. The above two methods are both analyzed in the next part.

## IV. MARINE EXPERIMENT

The experiment is conducted in the Indian Ocean. The purpose is to verify the stability and accuracy of the designed gravimeter for long-term operation and the correctness of the postprocessing method of ocean gravity data. The experimental equipment includes one set of differential GNSS and one set of the gravity-aided navigation system (including one set of gravimeters and one set of high-precision INS).

### A. Experimental Data

Experimental data includes three parts: gravity anomaly map, navigation trajectory, and gravity measurement data.

*1) Gravity Anomaly Map:* The Bureau Gravimetrique International (BGI) brings together the scientific services of the International Association of Geodesy (IAG), the International Gravity Field Service (IGFS), and the International Union of Geodesy and Geophysics (IUGG). The scientific services of BGI improve international understanding of the Earth's gravity field by collecting, homogenizing, and verifying all available gravity measurements (relative or absolute) and provide this information to a variety of users for scientific research. The gravity anomaly map of the gravity field of the Makran Trench in the North Indian Ocean is provided by BGI. The gravity anomaly map of the gravity field is shown in Fig. 9. The latitude range is from 23.5°N to 25.2°N, and the longitude range is from 61.8°E to 63.5°E, the resolution is $2' \times 2'$, and the gravity anomaly range is from $-90.58$ to 66.11 mGal.

*2) Navigation Trajectory:* Navigation trajectory is shown in Fig. 10. The sailing time is 3 h. The black line is the real trajectory obtained by the differential GNSS, and the blue line is the INS trajectory. The INS trajectory has an initial position error and two obvious Shura oscillation periods, and the error diverges with time.

Based on the real trajectory, the measured gravity anomaly values are compared with the gravity anomaly map, and the

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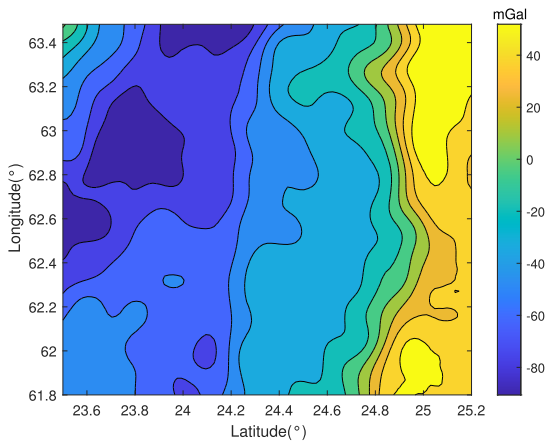


Fig. 9. Gravity anomaly map of the Makran Trench of the northern Indian Ocean.

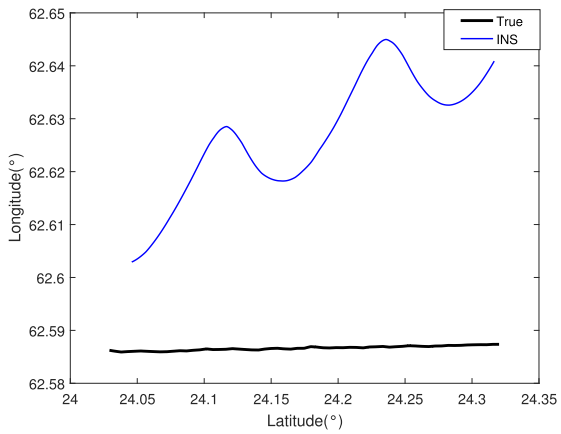


Fig. 10. Navigation trajectory.

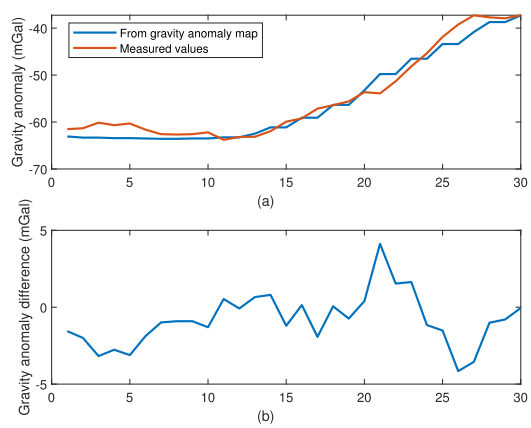


Fig. 11. Difference between measurement and gravity anomaly map. (a) Continuous sampling of gravity measurements. (b) Continuous sampling of gravity measurements.

results are shown in Fig. 11. It can be seen that the measurement error is slightly larger at the beginning of the trajectory, with a large deviation at 21 and 26. Most of the time the difference is less than 2 mGal.

*3) Real-Time Gravity Measurement Data:* The platform-type gravimeter used in the experiment is shown in Fig. 12.

Fig. 12. Platform gravimeter.

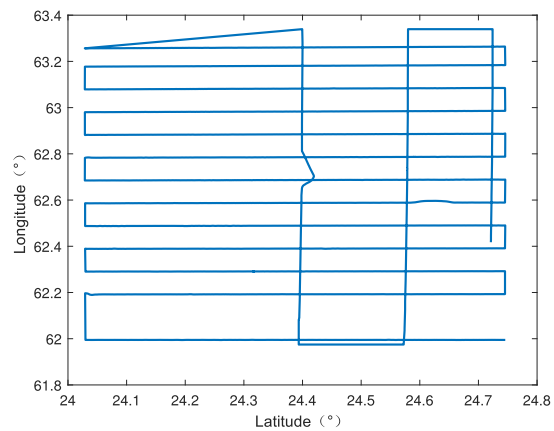


Fig. 13. Sixteen trajectories.

The measurement data used for gravity matching have been compensated for errors. Take zero-drift as an example. Gravimeters in related components affected by temperature change and material fatigue characteristics in static conditions on a fixed point measurement will have a slow drift with time, in view of the compensation for zero drift phenomenon that the drift correction. Generally, short-term drift is regarded as a linear change and is compensated by the results of precalibration measurement and postcalibration measurement performed at the same fixed point before and after the measurement operation. The compensated gravity anomaly value is calculated by

$$\delta g = -\frac{(f_b - f_a)}{(t_b - t_a)}(t - t_a) \tag{34}$$

where $f_a$ and $f_b$ are the mean of gravity of precalibration and postcalibration, respectively, $t_a$ and $t_b$ are the corresponding times, and $t$ is the time when the gravity is output.

The marine experiment provides 16 available trajectories which are shown in Fig. 13. We choose one of them to verify the proposed algorithm. Thirty-four cross-point discrepancies are counted. The cross-point discrepancy value is used as the evaluation index to test the performance of the gravimeter. The error is obtained according to these cross-point discrepancies which is shown in Table II.

From Table II, the root mean square of the gravimeter is 1.4511 mGal, the maximum error is 4.2 mGal, and the minimum error is −4.6 mGal. The results show that the gravimeter meets the experiment indicators.

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TABLE II
PERFORMANCE EVALUATION OF GRAVIMETER

| Performance Index | Error (mGal) |
|---|---|
| Root Mean Square | 1.4511 |
| Maximum Error | 4.2 |
| Minimum Error | -4.6 |

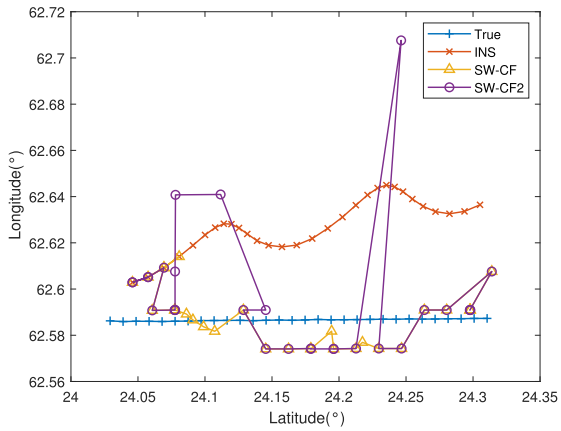


Fig. 14. Comparison between CFs with and without mismatch algorithm.

## B. Experimental Results

First, the function of the mismatch algorithm is verified and SW-CF is used. For SW-CF, the sequence length is $N = 3$ in Section III. In gravity anomaly data preprocessing, $m = 10$ and $l = 5$. In (22), $c_1 = c_2$. In (25) and (26), $T_a = 60°$ and $T_r = 4$. The matching period is 360 s. The result is shown in Fig. 14. The SW-CF is with mismatch algorithm, while SW-CF2 is without mismatch algorithm. It can be seen that mismatches mainly occur in two places. The reason for the occurrence of the first place is mainly because the trajectory is in the area of weak adaptation for a period of time in the beginning, which can be seen from Fig. 11(a). In the second place, there is a large deviation in the gravity measurement as shown in Fig. 11(b). SW-CF with mismatch algorithm successfully solves the problem of large error of matching result. Therefore, it can be proved that the mismatch matching method is effective.

Next, the localization results of the proposed algorithm will be evaluated. ICCP, SITAN, PF, PMF, mean square difference (MSD), outlier robust cubature Kalman filter (ORCKF) [22], and improved particle swarm optimization (IPSO) [23] algorithms are compared with two kinds of CF algorithms which both use mismatch algorithm in Section III. The INS trajectory and true trajectory are shown in Fig. 10. The initial position error of INS is 1.3543 nmi. ICCP algorithm adopts the sliding window method, and the matching sequence length is 6. The initial state of SITAN and ORCKF is $\mathbf{x}_0 = \mathbf{0}_{13\times 1}$ and initial covariance matrix is $\mathbf{P}_0 = \text{diag}([0.5'; 0.5'; 0.1 \text{ m/s}; 0.1 \text{ m/s}; 0.1°; 0.1°; 0.1°; 50\ \mu\text{g}; 50\ \mu\text{g}; 50\ \mu\text{g}; 0.001°/\text{h}; 0.001°/\text{h}; 0.001°/\text{h}]^2)$. The process noise covariance matrix is $\mathbf{Q}_k = \text{diag}([50\ \mu\text{g}; 50\ \mu\text{g}; 50\ \mu\text{g}; 0.001°/\text{h}; 0.0014°/\text{h}; 0.001°/\text{h}]^2)$ and the measurement error covariance matrix is $\mathbf{R}_k = \text{diag}(2 \text{ mGal})^2$. In ORCKF, $\nu_k = 5$, $N_{\max} = 30$, and $\epsilon = 10^{-6}$. The initial covariance matrix of PF and PMF are both $\text{diag}([0.5'; 0.5']^2)$. The process and measurement error covariance matrix are both $\text{diag}([0.3'; 0.3']^2)$ and $\text{diag}(2 \text{ mGal})^2$. The MSD method is applied in TERCOM, and the matching sequence length is 6. In IPSO, $c_1 \in [3, 5]$, $c_2 \in [4, 5]$ and $\omega \in [0.5, 1.5]$. IPSO is calculated in groups as shown in [23].

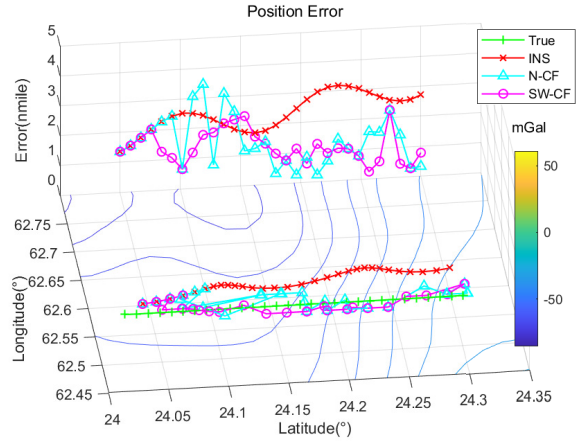


Fig. 15. Trajectories and position error of N-CF and SW-CF.

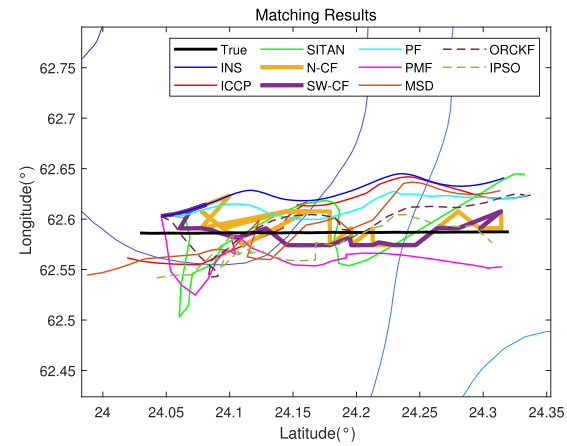


Fig. 16. Results of different methods.

For a clearer result, the trajectories and position error of N-CF and SW-CF are shown in Fig. 15. The matching results of the algorithms are shown in Fig. 16. From Fig. 16, we can see that due to the gravity measurement error and the initial error of the inertial navigation, a mismatch occurs in some algorithms. The N-CF and SW-CF algorithms sometimes failed to find the intersection due to the gravity measurement error. The position error is shown in Fig. 17. Affected by the initial error and measurement error, in most cases, the matching errors of ICCP, SITAN, PF, and PMF algorithms are greater than the inertial navigation error, while the N-CF and SW-CF matching errors have always been smaller than inertial navigation. The latitude and longitude errors are shown in Fig. 18. The latitude and longitude errors of the N-CF and SW-CF algorithms are smaller than other algorithms, and the fluctuation is smaller than others. From Table III, we can see that the average matching errors of ICCP, SITAN, PF,

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TABLE III
STATISTICS RESULTS OF CF AND OTHER MATCHING ALGORITHMS

| Method | Position error/nmile | | Absolute latitude error/′ | | Absolute longitude error/′ | |
|---|---|---|---|---|---|---|
| | Mean | RMSE | Mean | RMSE | Mean | RMSE |
| INS | 2.6115 | 2.6659 | 0.5806 | 0.7091 | 2.4918 | 2.5698 |
| ICCP | 2.4255 | 2.4891 | 1.0347 | 1.3085 | 1.9886 | 2.1175 |
| SITAN | 2.2781 | 2.5452 | 1.1532 | 1.3182 | 1.8013 | 2.1772 |
| N-CF | 1.4677 | 1.7203 | 1.0547 | 1.3674 | 0.8824 | 1.0439 |
| SW-CF | 1.2426 | 1.3750 | 1.0241 | 1.1987 | 0.5975 | 0.7380 |
| PF | 1.7147 | 1.7810 | 0.4208 | 0.5324 | 1.6232 | 1.6996 |
| PMF | 2.0360 | 2.1155 | 0.9601 | 1.1594 | 1.5856 | 1.7694 |
| MSD | 1.8044 | 2.0223 | 0.7159 | 1.0805 | 1.6750 | 1.8741 |
| ORCKF | 1.5055 | 1.6062 | 0.8752 | 0.9826 | 1.1748 | 1.3928 |
| IPSO | 1.2416 | 1.4478 | 0.6088 | 0.8807 | 1.0310 | 1.2588 |

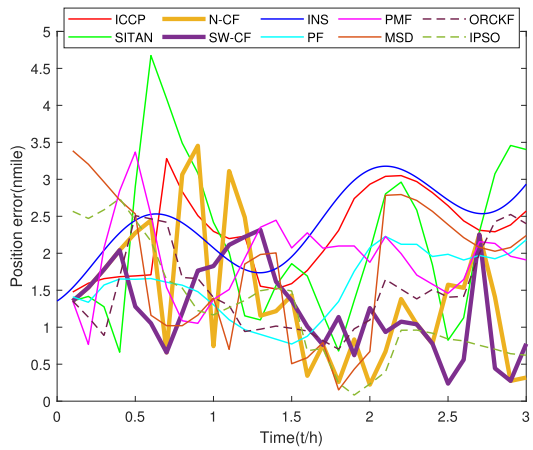


Fig. 17. Position error.

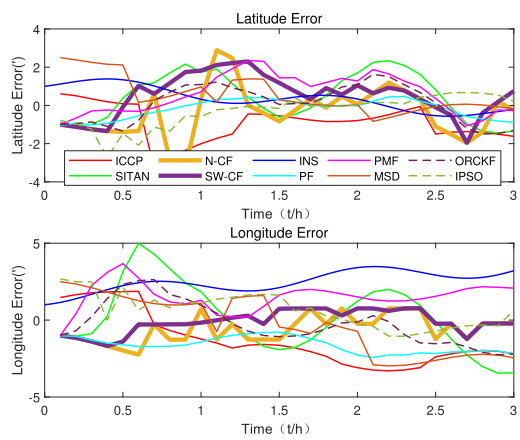


Fig. 18. Longitude and latitude error.

PMF, MSD, ORCKF, IPSO, N-CF, and SW-CF algorithms are 2.4255, 2.2781, 1.7147, 2.0360, 1.8044, 1.5055, 1.2416, 1.4677, and 1.2426 nmi. Compared with ICCP, SITAN, PF, PMF, MSD, and ORCKF, the accuracy of CF is improved by 40.60%, 35.57%, 14.40%, 27.91%, 18.66%, and 2.51%. SW-CF further improves the 15.34% accuracy of the N-CF algorithm. In the latest research, ORCKF is proposed for the problem of non-Gaussian heavy-tailed measurement noise which has a good performance as N-CF in this article. This also shows that the N-CF algorithm has a certain degree of robustness due to the data preprocessing and search range. IPSO has the same accuracy as SW-CF while it works badly at the beginning. The reason is that IPSO is affected by gravity measurement error. In summary, the SW-CF algorithm has the best performance. In addition, the continuity of SW-CF is better than that of the N-CF algorithm. The discussion of other algorithms is in the next part. Hence, the experimental results verify the effectiveness of the proposed two algorithms.

## V. DISCUSSION

In this part, the simulation results of this article and [3], [9], [10], [11], [13], [17], and [18] are discussed. The simulation in this article uses ICCP, SITAN, PF, PMF, MSD, N-CF, and SW-CF for comparison. The ICCP, SITAN, PF, PMF, and MSD methods have larger error when dealing with large initial position errors and gravity measurement errors. Wei et al. [3] is based on ICCP, Han et al. [9] and [17] are all matching algorithms based on PF, Wang et al. [10] and [11] are all matching algorithms based on PMF, and Han et al. [13] is an improved TERCOM-based matching algorithm.

Wang et al. [10] and [11] are based on PF. The PF requires high accuracy of the initial position of the INS. When the initial position error is large and unknown, the matching accuracy decreases or even diverges, making the matching results unable to meet the requirements. Besides, when calculating the posterior probability of particles appearing near the contour, the particles are similar in likelihood, and it is difficult for the filter to converge to the correct position. The factor of considering the initial position error is not mentioned in [10] and [11]. The initial position errors set by Wang et al. [10] and [11] are all 0 in the simulations, which means that the initial errors are not considered. Therefore, it can be inferred that the methods of [10] and [11] based on PF are not suitable for dealing with situation where the initial error is large.

Han et al. [18] is based on ICCP. The ICCP algorithm is affected by initial error and measurement error. Since the ICCP algorithm needs to find corresponding contour lines for rigid transformation, large measurement errors will affect the efficiency of the algorithm. The initial error will make the rigid transformation of inertial navigation trajectory points fail to converge to the optimal position, so the algorithm fails. In this article, initial error and measurement error both exist and measurement error even appears large deviation. Therefore, the shape of the ICCP trajectory is similar to the INS trajectory.

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Wei et al. [3] is based on SITAN. The SITAN algorithm is derived from terrain-matching navigation. It mainly linearizes the gravity background map locally, while it will bring a certain linearization error. In addition, since linearization needs to select a certain size area near the INS position for fitting, it is necessary to ensure that the real position is within the fitting area. When the INS error becomes larger and larger, first, it cannot be guaranteed that the real position is within the fitting area. And, second, it is easy to cause low fitting accuracy due to too large an adaptation area. After linearization, the measurement matrix is wrong which reduces the filter accuracy. In this article, the existence of an initial error will slow down the convergence speed of the filter.

Han et al. [9] and [17] are based on the PMF. The PMF is well applied in terrain navigation. However, in gravity-aided navigation, due to the high similarity of the gravity anomaly map, the accuracy of the initial point of the filter is high. When calculating the posterior probability, many points on the grid point have the same likelihood, and it is difficult for the filter to converge to the correct position. Complicated calculation of prior probability also increases the computational pressure. In [9] and [17], the initial position errors set by the simulation condition are $0.1'$, and the gyro drift is $0.01°$/h. For long-term underwater vehicles, the error can easily reach $1'$. Hence, the initial error $0.1'$ can be negligible. Therefore, it can be inferred that the methods of [9] and [17] based on the PMF are not suitable for dealing with situations with large initial errors.

The improved TERCOM-based algorithm in [13] only finds candidate points in the confidence interval generated by INS. When the inertial navigation error is large, too many sampling points will cause the matching speed to be slow. When the algorithm is working, only the information of the last point is used for matching. The rest of the information is discarded, and the utilization of observation information is low. The algorithm only considers the value of gravity anomaly for matching, when the gravity measurement is not accurate mismatch happens often. Besides the position information of the INS is not fully utilized.

Compared with [3], [9], [10], [11], [13], [17], and [18], the N-CF and SW-CF algorithms have better results. Other algorithms will inevitably have some problems when applied in GAINS. When these problems arise, the algorithms are easy to fail. For example, there are some problems such as the gravity measurement error, inertial navigation initial error, and gravity anomaly map similarity. The algorithms proposed in this article consider the above problems at the same time, so they have good results. For gravity measurement errors, this article presents the gravity data-preprocessing method and search range. The initial error of INS is avoided by CF in the search range (this is because CF does not have high requirements on the initial position error of INS during calculation). For similarity of gravity anomaly maps, the INS template within CF is added to avoid this problem In addition, the use of the mismatch algorithm makes the matching result more accurate. To solve the real-time problem, SW-CF is proposed. SW-CF has all the above characteristics. It not only has real-time performance, but also has more continuous results and higher accuracy compared with the N-CF algorithm.

## VI. Conclusion

This article presents a matching algorithm using CF. the CF algorithm is divided into gravity data preprocessing, determining the search range, determining the filter template, CF, and mismatch detection. Through these steps, the CF matching algorithm makes full use of the gravity measurement data and the inertial navigation information and effectively overcomes the influence of the initial error and gravity measurement error. In order to meet the real-time requirements, SW-CF is proposed. The simulation results show that CF and SW-CF both have an effect on both initial errors and measurement errors. Since the algorithm uses the traversal method when searching, the next step is to optimize the algorithm to improve the speed.

## Acknowledgment

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## References

[1] L. Medagoda, S. B. Williams, O. Pizarro, J. C. Kinsey, and M. V. Jakuba, "Mid-water current aided localization for autonomous underwater vehicles," *Auto. Robots*, vol. 40, no. 7, pp. 1207–1227, Oct. 2016.

[2] S. Zhao, L. Shi, W. Zhang, and Z. Deng, "Global dynamic path-planning algorithm in gravity-aided inertial navigation system," *IET Signal Process.*, vol. 15, no. 8, pp. 510–520, Oct. 2021.

[3] E. Wei et al., "A robust solution of integrated SITAN with TERCOM algorithm: Weight-reducing iteration technique for underwater vehicles' gravity-aided inertial navigation system," *Navigation*, vol. 64, no. 1, pp. 111–122, 2017.

[4] M. Liu, B. Wang, Z. Deng, and M. Fu, "Improved ICCP algorithm and its application in gravity matching aided inertial navigation system," in *Proc. 33rd Chin. Control Conf.*, Nanjing, China, Jul. 2014, pp. 562–567.

[5] H. Zhang, L. Yang, and M. Li, "Improved ICCP algorithm considering scale error for underwater geomagnetic aided inertial navigation," *Math. Problems Eng.*, vol. 2019, pp. 1–9, Apr. 2019.

[6] Y. Han, B. Wang, Z. Deng, S. Wang, and M. Fu, "A mismatch diagnostic method for TERCOM-based underwater gravity-aided navigation," *IEEE Sensors J.*, vol. 17, no. 9, pp. 2880–2888, May 2017.

[7] L. Hostetler and R. Andreas, "Nonlinear Kalman filtering techniques for terrain-aided navigation," *IEEE Trans. Autom. Control*, vol. AC-28, no. 3, pp. 315–323, Mar. 1983.

[8] Y. Chen, H. Li, Z. Qiu, T. D. Wang, and K. R. Oldham, "Improved extended Kalman filter estimation using threshold signal detection with an MEMS electrostatic microscanner," *IEEE Trans. Ind. Electron.*, vol. 67, no. 2, pp. 1328–1336, Feb. 2020.

[9] Y. Han, B. Wang, Z. Deng, and M. Fu, "A matching algorithm based on the nonlinear filter and similarity transformation for gravity-aided underwater navigation," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 2, pp. 646–654, Apr. 2018.

[10] B. Wang, J. Zhu, Z. Deng, and M. Fu, "A characteristic parameter matching algorithm for gravity-aided navigation of underwater vehicles," *IEEE Trans. Ind. Electron.*, vol. 66, no. 2, pp. 1203–1212, Feb. 2019.

[11] B. Wang, L. Yu, Z. Deng, and M. Fu, "A particle filter-based matching algorithm with gravity sample vector for underwater gravity aided navigation," *IEEE/ASME Trans. Mechatronics*, vol. 21, no. 3, pp. 1399–1408, Jun. 2016.

[12] L. Wu, H. Wang, H. Chai, H. Hsu, and Y. Wang, "Research on the relative positions-constrained pattern matching method for underwater gravity-aided inertial navigation," *J. Navigat.*, vol. 68, no. 5, pp. 937–950, Sep. 2015.

[13] Y. Han, B. Wang, Z. Deng, and M. Fu, "An improved TERCOM-based algorithm for gravity-aided navigation," *IEEE Sensors J.*, vol. 16, no. 8, pp. 2537–2544, Apr. 2016.

[14] D. Tan, M. Lingjuan, S. Haijun, and S. Yongsheng, "Solving gravity anomaly matching problem under large initial errors in gravity aided navigation by using an affine transformation based artificial bee colony algorithm," *Frontiers Neurorobot.*, vol. 13, p. 19, May 2019.

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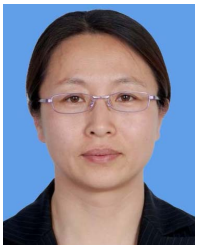
[15] Z. Yang, Z. Zhu, and W. Zhao, "A triangle matching algorithm for gravity-aided navigation for underwater vehicles," *J. Navigat.*, vol. 67, no. 2, pp. 227–247, Mar. 2014.
[16] Z. Zhuangsheng, G. Yiyang, and Y. Zhenli, "Study on initial gravity map matching technique based on triangle constraint model," *J. Navigat.*, vol. 69, no. 2, pp. 353–372, Mar. 2016.
[17] Y. Han, B. Wang, Z. Deng, and M. Fu, "Point mass filter based matching algorithm in gravity aided underwater navigation," *J. Syst. Eng. Electron.*, vol. 29, no. 1, pp. 152–159, Feb. 2018.
[18] Y. Han, B. Wang, Z. Deng, and M. Fu, "A combined matching algorithm for underwater gravity-aided navigation," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 1, pp. 233–241, Feb. 2018.
[19] M. Huang, *Measurement and Application of Marine Gravity Field*. Beijing, China: Surveying and Mapping Press, 2005.
[20] D. Bolme, J. R. Beveridge, B. A. Draper, and Y. M. Lui, "Visual object tracking using adaptive correlation filters," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2544–2550.
[21] H. W. Park, J. G. Lee, and C. G. Park, "Covariance analysis of strapdown INS considering gyrocompass characteristics," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 31, no. 1, pp. 320–328, Jan. 1995.
[22] Z. Wang, Y. Huang, M. Wang, J. Wu, and Y. Zhang, "A computationally efficient outlier-robust cubature Kalman filter for underwater gravity matching navigation," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–18, 2022.
[23] J. Zou and T. Cai, "Improved particle swarm optimization screening iterative algorithm in gravity matching navigation," *IEEE Sensors J.*, vol. 22, no. 21, pp. 20866–20876, Nov. 2022.

**Shengwu Zhao** received the B.S. degree from the Beijing Institute of Technology, Beijing, China, in 2020, where he is currently pursuing the Ph.D. degree.

His current research interests include strapdown inertial navigation systems, integrated navigation, and information fusion.

**Xuan Xiao** received the B.S. degree in automation and the M.S. and Ph.D. degrees from the Beijing Institute of Technology, Beijing, China, in 1997, 2000, and 2011, respectively.

He has been teaching at the Beijing Institute of Technology since 2000. He worked as a Visiting Scholar with the University of Victoria, Footscray, VIC, Australia, in 2012. He is currently an Associate Professor with the Integrated Navigation and Intelligent Navigation Laboratory, School of Automation, Beijing Institute of Technology. His research interests include advanced technology of sensing and detecting and integrated navigation of unmanned ground vehicles.



**Zhihong Deng** received the B.S. and Ph.D. degrees in control theory and engineering from the Department of Automation, Harbin Engineering University, Harbin, China, in 1997 and 2001, respectively.

From 2001 to 2003, she was a Postdoctoral Researcher with the Beijing Institute of Technology, Beijing, China. Since 2004, she has been with the Department of Automatic Control, Beijing Institute of Technology, where she was first an Associate Professor and has been a Professor since 2010. Her current research interests include strapdown inertial navigation systems, information fusion and filter, and intelligent navigation technology.

**Lei Shi** received the B.S. degree from the School of Automation, Beijing Institute of Technology, Beijing, China, in 2015, where he is currently pursuing the Ph.D. degree.

His current research interests include strapdown inertial navigation systems, information fusion and filter, and intelligent navigation technology.

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